

Lung-Region Focused Deep Learning Framework for Multi-Class Pneumonia Detection Using EfficientNetV2 and Confidence-Aware Decision Support

Dissertation Report

Submitted in partial fulfilment of the requirement of Master of
Science in Data Science

By

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2024 - 2026

DECLARATION

I undersigned hereby declare that the project report “**Lung-Region Focused Deep Learning Framework for Multi-Class Pneumonia Detection Using EfficientNetV2 and Confidence-Aware Decision Support**”, submitted for partial fulfilment of the requirements for the award of the degree of Master of Science in Data Science, Kerala University, is a Bonafide work done by me under the supervision of Rahul D, Assistant Professor, Department of Futures Studies.

This submission represents my original work and where ideas or words of others have been included, they have been duly acknowledged, cited, and referenced. I further declare that I have adhered to the principles of academic honesty and integrity and have not misrepresented or fabricated any data, source, or result in this report.

I also understand that any violation of the above will lead to disciplinary action by the Institute/University and may invoke penal action from the concerned authorities. This report has not been previously submitted for the award of any degree, diploma, or title at any other University or Institution.

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CERTIFICATE

This is to certify that the report entitled “**Lung-Region Focused Deep Learning Framework for Multi-Class Pneumonia Detection Using EfficientNetV2 and Confidence-Aware Decision Support**”, submitted by **RIFA FATHIMA** to the University of Kerala, Karyavattom in partial fulfilment of the requirements for the award of the Degree of **Master of Science in Data Science, Department of Futures Studies** is a Bonafide record of the project work carried out by her under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ACKNOWLEDGEMENT

I would like to express heartfelt thanks to my guide assistant professor Rahul D, Department of Futures Studies, University of Kerala for their help, advice, support, constant encouragement and valuable guidance. I would like to thank faculties from the Department of Futures Studies Dr. Thara Prabhakar, Dr. Susan George and all my friends, family, whose continued support and inspiration helped me to accomplish this task.

RIFA FATHIMA

ABSTRACT

Pneumonia remains one of the leading causes of respiratory illness worldwide, which require timely and accurate diagnosis to reduce morbidity and mortality. Chest X-ray imaging is widely used as a primary diagnostic tool due to its accessibility and cost-effectiveness; however, manual interpretation is time-consuming and prone to inter-observer variability. Recent advances in deep learning have enabled automated analysis of radiographic images, offering potential support for clinical decision-making. Despite promising results, many existing approaches rely solely on classification accuracy.

This project presents a lung-region focused deep learning framework for multi-class pneumonia detection from chest X-ray images. The proposed system integrates U-Net-based lung segmentation to isolate pulmonary regions and suppress irrelevant background structures, ensuring that the classification model focuses on clinically meaningful areas. Contrast enhancement and preprocessing techniques are applied to improve the visibility of subtle radiographic patterns such as ground-glass opacities and infiltrates.

EfficientNetV2-B0 is employed as the primary classification backbone due to its efficient compound scaling strategy and superior training stability. A two-stage transfer learning approach is adopted, consisting of classifier head training followed by selective fine-tuning of higher-level layers to adapt pretrained features to pneumonia-specific characteristics. To enhance robustness and generalization, data augmentation techniques are applied during training.

Beyond traditional evaluation metrics, this study incorporates model interpretability and reliability assessment and confidence calibration via temperature scaling. A confidence-based decision support mechanism is introduced to guide automated acceptance, expert review, or further diagnostic investigation based on prediction reliability.

Experimental results demonstrate that lung-aware preprocessing and confidence calibration improve the trustworthiness and clinical relevance of automated pneumonia detection systems. Although segmentation and calibration introduce conservative prediction behaviour, they reduce shortcut learning and overconfidence, contributing to safer and more reliable diagnostic support.

The proposed framework emphasizes reliability, interpretability and clinical applicability, highlighting the importance of integrating explainable AI and confidence-aware decision mechanisms in medical imaging systems. This approach provides a foundation for future research in lesion localization and multimodal clinical integration.

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INTRODUCTION

Respiratory diseases, particularly pneumonia, continue to pose a significant global health burden. Pneumonia is an inflammatory condition of the lung tissue caused by bacterial, viral, or fungal infections, and its early detection is critical for effective treatment and prevention of severe complications. Chest radiography remains the most commonly used imaging modality for pneumonia diagnosis due to its affordability, rapid acquisition, and widespread availability. However, interpreting chest X-ray images requires expert radiological knowledge and is subject to human error, fatigue, and inter-observer variability.

Numerous studies have applied deep learning techniques to pneumonia detection, reporting high classification accuracy. Nevertheless, many of these approaches rely on end-to-end classification without addressing the reliability and interpretability of model predictions. Additionally, models trained on raw chest X-rays may inadvertently learn irrelevant background features such as image borders, medical device artifacts, or text annotations, leading to shortcut learning and reduced clinical trustworthiness.

To address these limitations, lung-region focused preprocessing has emerged as an effective strategy for guiding models toward anatomically relevant areas. Lung segmentation methods, particularly U-Net architectures, enable accurate extraction of pulmonary regions from chest radiographs. This segmentation ensures that the classification model learns disease-relevant features, improving robustness and interpretability. Image enhancement techniques can improve contrast and visibility, allowing deep learning models to better capture pathological features. Combining lung-focused preprocessing with enhancement techniques provides a more clinically meaningful input for classification models.

EfficientNetV2 has emerged as a modern CNN architecture that achieves high performance with reduced computational cost through compound scaling and optimized convolutional blocks. Its ability to learn rich feature representations while maintaining training efficiency makes it well-suited for medical imaging applications. Transfer learning further enhances performance by leveraging pretrained knowledge from large-scale datasets and adapting it to domain-specific tasks.

Despite advances in classification accuracy, clinical adoption of AI-based diagnostic systems requires reliability and transparency. Neural networks are known to produce overconfident predictions, which may be unsafe in healthcare settings. Confidence calibration techniques adjust predicted probabilities to better reflect true likelihoods, enabling risk-aware decision-making. Additionally, explainability methods such as occlusion sensitivity analysis and feature visualization provide insights into model behaviour, supporting clinical trust.

This project proposes an integrated framework for pneumonia detection that combines lung segmentation, image enhancement, EfficientNetV2-based classification, confidence calibration, and explainability analysis. By emphasizing reliability, interpretability, and clinically meaningful preprocessing, the proposed system aims to move beyond accuracy-centric approaches and contribute toward safer AI-assisted pneumonia screening.

LITERATURE REVIEW

The application of deep learning techniques for pneumonia detection from chest X-ray images has attracted significant research interest in recent years. Early approaches primarily focused on improving classification accuracy using convolutional neural networks (CNNs). However, more recent studies emphasize preprocessing, segmentation, explainability, and reliability to enhance clinical applicability. The reviewed literature can be broadly categorized into the following conceptual areas:

1. Deep learning architectures for pneumonia detection
2. Lung segmentation and region-focused learning
3. Image enhancement and preprocessing techniques
4. Explainability and visualization methods
5. Confidence calibration and reliability in medical AI

Base Paper: Multi-Class Pneumonia Detection Using Transfer Learning

The base paper by Ibrahim et al. (2021) presents a deep learning framework for classifying chest X-ray images into multiple categories: Normal, Bacterial Pneumonia, Viral Pneumonia, and COVID-19 Pneumonia. The study employs a transfer learning approach using the AlexNet architecture, which was pretrained on large-scale image datasets and fine-tuned for pneumonia classification.

Methodology:

The authors utilized chest X-ray images collected from publicly available repositories. Images were resized and normalized before being input into the AlexNet-based model. Transfer learning was applied to leverage pretrained weights, enabling efficient feature extraction despite limited medical imaging data. The classification task included one-way, two-way, three-way, and four-way classification scenarios. The four-class classification provided the most comprehensive evaluation of pneumonia subtypes.

Performance:

The model achieved an overall accuracy of approximately 92% in multi-class classification. The results demonstrated that transfer learning could effectively distinguish pneumonia types from chest X-ray images.

Strengths:

- Demonstrated feasibility of multi-class pneumonia classification
- Efficient use of transfer learning with limited data
- Evaluated multiple classification scenarios

Limitations:

Despite promising performance, several limitations were identified:

- AlexNet is a shallow architecture compared to modern CNNs
- No lung segmentation to guide attention toward relevant regions
- Absence of image enhancement to improve subtle pattern visibility
- Lack of explainability and interpretability methods
- No confidence calibration or reliability analysis
- Potential reliance on background artifacts

These limitations highlight opportunities for improvement through modern architectures, lung-focused preprocessing, and reliability-aware evaluation, aspects addressed in the present work.

1. Deep Learning Architectures for Pneumonia Detection

Deep convolutional neural networks have demonstrated strong performance in detecting pulmonary diseases from radiographic images. Rajpurkar et al. (2017) introduced **CheXNet**, a DenseNet-121-based model trained on the ChestX-ray14 dataset. The study reported radiologist-level performance in pneumonia detection with an accuracy of approximately 0.76. This work demonstrated the feasibility of automated chest X-ray interpretation and inspired subsequent deep learning research in medical imaging.

Tan and Le (2019) proposed **EfficientNet**, a family of CNN architectures that scale network depth, width, and resolution using a compound scaling strategy. EfficientNet models achieved state-of-the-art performance on ImageNet while maintaining computational efficiency. Although not designed specifically for medical imaging, EfficientNet's efficiency and feature representation capability make it highly suitable for healthcare applications.

Banerjee & Monir (2024) showed that EfficientNetV2 outperforms V1 in medical imaging tasks, provide faster convergence and improved stability.

Chowdhury et al. (2020) applied transfer learning-based CNN architectures to detect COVID-19 and viral pneumonia from chest X-rays. Their results showed accuracy levels between 90% and 95%, demonstrating the potential of deep learning for distinguishing respiratory infections. However, the study highlighted challenges in differentiating viral pneumonia types due to overlapping radiographic features. In addition, Moses (2021) presented a detailed review on deep learning techniques applied to automatic disease detection in chest X-rays.

2. Lung segmentation and region-focused learning

While classification-based approaches focus on predicting disease labels, segmentation-based methods aim to identify regions of interest within medical images. Ronneberger et al. introduced U-Net, a convolutional neural network designed specifically for biomedical image segmentation. U-Net architectures, have shown exceptional performance in extracting anatomical structures. Haghanifar et al. (2022) introduced a lung-focused preprocessing approach for COVID-19 detection, demonstrating that segmentation improves classification

performance by removing irrelevant background structures. By guiding the model's attention to pulmonary regions, segmentation reduces noise and shortcut learning.

3. Image enhancement and preprocessing techniques

Chest X-ray images often suffer from low contrast and poor visibility of pathological features. Image enhancement methods can improve feature clarity and assist deep learning models in detecting subtle abnormalities.

Altan and Narlı (2022) demonstrated that applying **Contrast Limited Adaptive Histogram Equalization (CLAHE)** significantly improved COVID-19 detection accuracy. Enhancement techniques improve the visibility of ground-glass opacities and lung infiltrates, which are essential indicators of pneumonia.

4. Explainability and visualization methods

Explainability is a critical requirement for clinical acceptance of AI-based diagnostic systems. Simonyan et al. (2014) introduced occlusion sensitivity analysis as a model-agnostic approach to visualize CNN decision-making by systematically masking parts of the input image. While gradient-based methods such as Grad-CAM are commonly used, their effectiveness diminishes in very deep architectures like EfficientNetV2 due to gradient diffusion. Occlusion-based explainability, on the other hand, provides stable and interpretable visual explanations by directly measuring the impact of image regions on model predictions. This makes occlusion sensitivity particularly suitable for explaining predictions in deep medical imaging models.

5. Confidence calibration and reliability in medical AI

Modern deep neural networks are often overconfident in their predictions, which poses serious risks in clinical applications. Guo et al. (2017) systematically analysed the calibration of neural networks and proposed temperature scaling as an effective post-hoc calibration technique that improves the reliability of predicted confidence scores without altering classification accuracy. In medical imaging, poorly calibrated confidence scores can lead to unsafe diagnostic decisions. Moses (2021) emphasized the necessity of confidence-aware systems in clinical AI pipelines. Reliable probability estimates enable risk-aware decision support systems, allowing uncertain cases to be flagged for expert review.

Research Gaps and Advancements of the Proposed Work

Limitations in Existing Studies	Advancements in the Proposed Work
Many studies focus solely on classification accuracy without considering reliability.	Incorporates confidence calibration to improve reliability of probability estimates.
Models often learn from irrelevant background artifacts, leading to shortcut learning.	Integrates U-Net–based lung segmentation to ensure lung-region–focused learning.
Limited integration of segmentation with classification frameworks.	Combines lung segmentation with EfficientNetV2 classification for anatomically guided learning.
Subtle pneumonia patterns may be missed due to low image contrast.	Applies image enhancement techniques to improve visibility of faint pathological features.
Traditional CNN architectures may lack efficiency and training stability.	Employs EfficientNetV2-B0 for efficient feature extraction and improved training stability.
Lack of explainability reduces clinical trust in AI predictions.	Uses occlusion sensitivity and feature visualization to improve interpretability.
Deep learning models often produce overconfident predictions.	Applies temperature scaling to calibrate prediction confidence.
Existing systems provide predictions without reliability-based guidance.	Introduces confidence-based decision support for safer clinical interpretation.
Minimal emphasis on clinical applicability and workflow integration.	Designs a clinically meaningful pipeline supporting expert review and further diagnostics.

Problem Statement

Pneumonia remains a major global health concern, affecting millions of individuals annually and contributing significantly to morbidity and mortality. Early and accurate diagnosis is essential for effective treatment and prevention of severe complications. Chest X-ray imaging is widely used as the primary diagnostic tool due to its availability, affordability, and rapid acquisition. However, manual interpretation of chest radiographs requires specialized expertise and is subject to variability, fatigue, and human error.

Recent advances in deep learning have enabled automated pneumonia detection from chest X-ray images with promising accuracy. Despite these advancements, several challenges limit the clinical adoption of existing AI-based systems. Many models rely solely on classification accuracy and often learn spurious correlations from background artifacts such as text annotations, medical devices, or image borders. This phenomenon, known as shortcut learning, reduces model reliability and clinical trustworthiness.

Furthermore, chest X-ray images frequently exhibit low contrast and subtle pathological patterns, making disease detection difficult even for experienced radiologists. Without appropriate preprocessing and enhancement, deep learning models may fail to capture faint abnormalities such as ground-glass opacities and fine infiltrates.

Another critical challenge is the lack of reliability assessment in many AI systems. Neural networks often produce overconfident predictions, which may lead to unsafe clinical decisions. Most existing approaches do not provide confidence-aware decision support, interpretability, or mechanisms to flag uncertain cases for expert review.

Therefore, there is a need for a robust, interpretable, and reliability-aware pneumonia detection framework that focuses on clinically relevant lung regions, enhances subtle disease patterns, and provides confidence-based decision support to assist medical professionals.

OBJECTIVES OF THE STUDY

The primary goal of this project is to develop a reliable and interpretable deep learning framework for multi-class pneumonia detection from chest X-ray images.

Specific Objectives:

1. To develop a multi-class pneumonia classification system capable of distinguishing between Normal, Bacterial Pneumonia, Viral Pneumonia, and COVID-19 Pneumonia.
2. To implement lung-region segmentation using U-Net to isolate pulmonary regions and suppress irrelevant background structures.
3. To enhance chest X-ray images using preprocessing techniques to improve the visibility of subtle pathological features.
4. To employ EfficientNetV2-B0 as a modern deep learning architecture for efficient and accurate feature extraction.
5. To implement a three-stage transfer learning strategy involving classifier head training and fine-tuning for improved domain adaptation.
6. To evaluate model performance using comprehensive metrics, including precision, recall, F1-score, confusion matrix, and ROC analysis.
7. To incorporate explainability techniques such as occlusion sensitivity and feature visualization to improve model transparency.
8. To apply confidence calibration techniques to improve reliability of predicted probabilities.
9. To design a confidence-based decision support mechanism to guide automated acceptance, expert review, or further diagnostic investigation.
10. To develop a clinically meaningful AI-assisted framework that prioritizes reliability, interpretability, and safety alongside classification performance.

SYSTEM OVERVIEW AND PROPOSED FRAMEWORK

System Overview

The proposed system is an end-to-end deep learning framework designed to detect pneumonia from chest X-ray images. The system integrates preprocessing, lung segmentation, feature extraction, classification, and reliability assessment modules to ensure accurate and clinically meaningful predictions.

The workflow begins with the acquisition of chest X-ray images, followed by preprocessing steps including cleaning, resizing, normalization, and dataset balancing. Lung segmentation is then performed using a U-Net model to extract pulmonary regions and suppress irrelevant background structures. The lung-focused images are enhanced to improve contrast and highlight subtle disease patterns.

The processed images are subsequently fed into the EfficientNetV2-B0 deep learning model, which performs feature extraction and classification. The model is trained using a three-stage transfer learning strategy to ensure stable learning and effective adaptation to pneumonia-specific patterns.

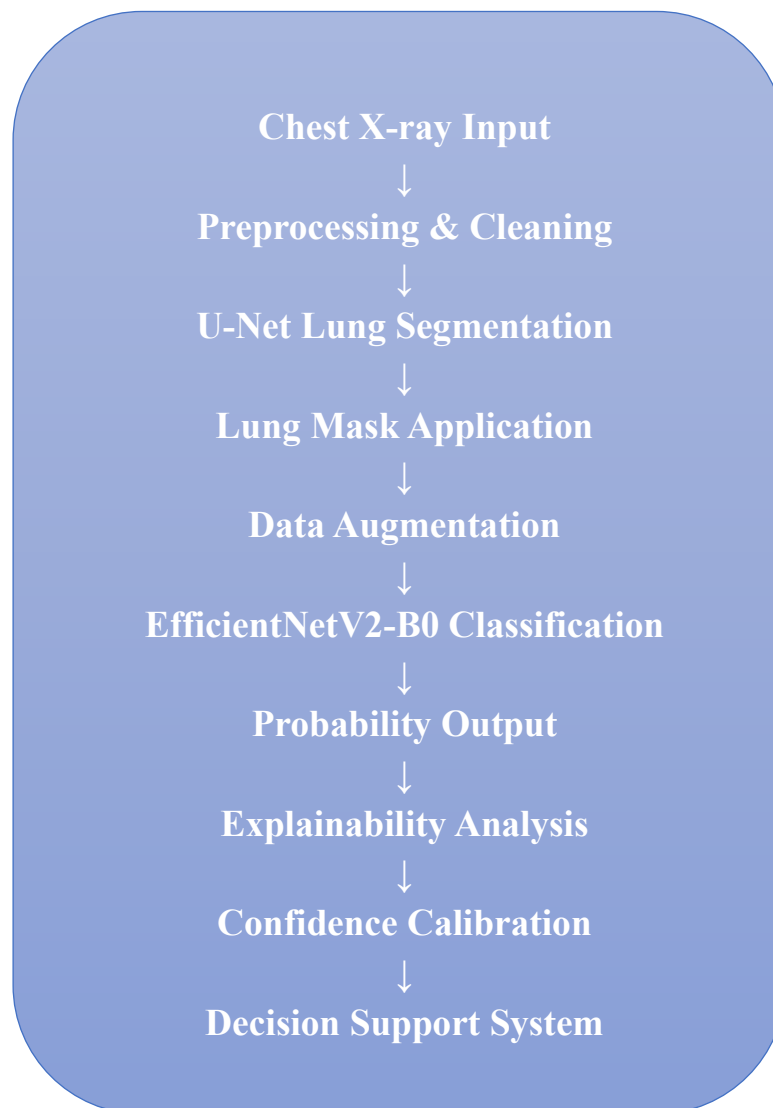
Following classification, the system evaluates prediction reliability using confidence calibration and provides decision support based on confidence thresholds. Explainability methods are incorporated to analyze model behavior and ensure predictions are based on clinically relevant regions.

Proposed Framework

The proposed framework introduces several enhancements to improve reliability, interpretability, and clinical applicability compared to conventional pneumonia detection systems.

- **Lung-Region Focused Learning:**
U-Net-based segmentation isolates lung regions before classification. This ensures the model focuses on anatomically relevant structures and reduces bias caused by background artifacts.
- **Image Enhancement for Feature Visibility:**
Contrast enhancement techniques improve the visibility of subtle radiographic features, enabling better detection of faint pneumonia patterns.
- **Efficient Feature Extraction Using EfficientNetV2-B0:**
EfficientNetV2-B0 is employed for its compound scaling and training efficiency, enabling high-performance feature extraction with reduced computational cost.

- **Two-Stage Transfer Learning Strategy:**
The training process involves classifier head training followed by fine-tuning of upper layers, ensuring stable convergence and domain adaptation.
- **Explainability and Feature Analysis:**
Occlusion sensitivity analysis and feature visualization techniques are used to interpret model decisions and verify that predictions rely on lung regions.
- **Confidence Calibration and Decision Support:**
Temperature scaling improves confidence reliability, and a confidence-based decision support mechanism assists clinicians by flagging uncertain cases.



The integrated pipeline ensures that predictions are not only accurate but also reliable and clinically meaningful. By combining lung-focused preprocessing, modern deep learning architecture, interpretability analysis, and confidence-aware decision support, the proposed framework addresses key limitations of existing pneumonia detection systems.

METHODOLOGY AND SYSTEM ARCHITECTURE

The proposed pneumonia detection system follows a structured deep learning pipeline designed to improve diagnostic reliability, interpretability, and clinical relevance. The methodology integrates data preprocessing, lung-region segmentation, image enhancement, deep feature extraction, classification, and reliability assessment.

1. Data Acquisition and Dataset Description

The dataset used in this study consists of chest X-ray images categorized into four classes:

- Normal: 20,384 images
- Bacterial Pneumonia: 12,024 images
- Viral Pneumonia: 2,690 images
- COVID-19 Pneumonia: 7,232 images

Images were collected from publicly available radiographic repositories and organized into class-specific directories. Each image represents a frontal chest radiograph captured under varying clinical conditions.

Since chest X-rays are grayscale images, they provide structural information of lung anatomy and pathological patterns essential for pneumonia diagnosis.

2. Data Preprocessing & Balancing

Preprocessing is essential to ensure dataset consistency, remove noise and improve model learning.

i. **Dataset Organization and Class Mapping**

The dataset paths are explicitly defined, and class labels are mapped to their corresponding directory names. Each image is assigned a class label based on its folder structure, enabling automatic label generation during dataset loading.

ii. **Data Cleaning**

Corrupted, unreadable, or duplicate images were removed to ensure dataset integrity and prevent training instability. Images with incompatible dimensions or invalid pixel values are excluded to ensure dataset consistency.

iii. **Image Resizing**

All images were resized to **224 × 224 × 3 pixels** to match the input requirements of EfficientNetV2-B0.

$$I_{resized} = \text{resize}(I_{original}, 224 \times 224)$$

Resizing ensures uniform input dimensions and efficient computation.

Although chest X-rays are grayscale, they are converted to 3 channels to match ImageNet-pretrained CNN requirements and to leverage pretrained EfficientNetV2 weights because using pretrained weights drastically reduces training time and improves generalization on limited medical datasets.

iv. Pixel Normalization

Pixel values were scaled to the range [0,1] to stabilize gradient updates during training:

$$I_{norm} = \frac{I}{255}$$

Normalization improves convergence speed and training stability.

v. Dataset Balancing

Class imbalance can bias model predictions. Controlled resampling techniques like oversampling and under sampling were applied to achieve balanced class representation.

Balancing ensures:

- improved sensitivity across classes
- reduced bias toward dominant classes
- fair evaluation performance

3. Dataset Splitting

The preprocessed dataset is split into training, validation, and testing subsets using a stratified approach to preserve class distribution across splits.

- ✚ **Training set** — model learning
- ✚ **Validation set** — hyperparameter tuning
- ✚ **Test set** — final evaluation

4. Lung Focused Enhancement Using U-Net

Chest X-rays contain background structures such as ribs, labels, and external artifacts that may mislead deep learning models. Lung segmentation ensures that classification is performed only on anatomically relevant regions.

i. U-Net Architecture

A U-Net architecture was implemented for lung segmentation. U-Net consists of:

- Encoder path (feature extraction)
- Decoder path (spatial reconstruction)
- Skip connections (retain spatial detail)

The model performs pixel-wise binary classification:

- Foreground (1): Lung region
- Background (0): Non-lung region

ii. Loss Function

Binary Cross-Entropy (BCE) loss was used:

$$Loss = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

BCE is suitable for binary segmentation tasks.

iii. Lung Mask Application

The trained U-Net generates a lung mask for each input image. This mask is then multiplied pixel-wise with the original chest X-ray image. The predicted lung mask $M(x, y)$ is multiplied with the original image:

$$I_{lung}(x, y) = I(x, y) \times M(x, y)$$

This operation suppresses irrelevant background regions and retains pulmonary structures.

iv. Image Enhancement

Chest X-rays often exhibit low contrast and subtle pathological patterns. Image enhancement improves visibility of pneumonia-related features such as ground-glass opacities and infiltrates.

Enhancement techniques applied include:

- ✚ contrast adjustment
- ✚ brightness normalization
- ✚ subtle feature amplification

This improves feature clarity for deep learning models.

5. Data Augmentation

To improve model generalization and prevent overfitting, augmentation was applied to training images.

Augmentation techniques include:

- horizontal flipping
- small rotations
- zoom variations
- contrast perturbations

Augmentation simulates real-world imaging variations and increases data diversity.

6. EfficientNetV2-B0 for Feature Extraction

EfficientNetV2-B0 is used as the primary feature extractor due to its improved training stability over EfficientNetV1, efficient compound scaling strategy, faster convergence, better parameter efficiency and balanced depth, width, and resolution.

It extracts hierarchical feature representations like edges and textures in early layers, lung patterns and opacities in middle layers and disease-specific representations in deep layers.

i. Architecture of EfficientNetV2-B0

Core Building Blocks:

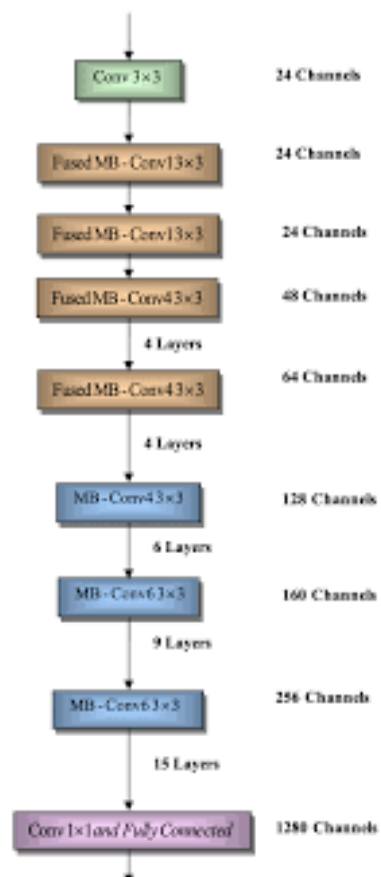
EfficientNetV2-B0 is built using two main types of convolution blocks:

1. Fused-MBConv Blocks (Early Layers)

- Combine convolution and expansion operations into a single block
- Improve training speed and stability
- Capture low-level features such as edges and textures

2. MBConv Blocks (Deeper Layers)

- Use depthwise separable convolutions
- Reduce parameter count and computation
- Capture high-level semantic features related to disease patterns



ii. Compound Scaling

EfficientNet scales network dimensions using:

- depth (α)
- width (β)
- resolution (γ)

This balanced scaling improves performance without excessive computation.

iii. Working Mechanism

1. Input Processing

The input chest X-ray image is resized to 224×224 and passed into the network. Although X-ray images are grayscale, they are replicated across three channels to match pretrained ImageNet weights.

2. Feature Extraction

The convolutional blocks progressively extract hierarchical features:

- Early layers learn edges and lung textures
- Middle layers learn structural lung patterns
- Deeper layers learn disease-specific representations

3. Global Average Pooling

Global Average Pooling (GAP) converts feature maps of size $H \times W \times C$ into a C -dimensional vector:

$$GAP_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W F_{i,j,c}$$

GAP reduces parameters, prevents overfitting, and improves generalization.

4. Classification Head

The classification head consists of:

- Global Average Pooling
- Dropout layer (regularization)
- Dense layer with Softmax activation

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

Softmax outputs probability distribution across the four pneumonia classes.

7. Training Strategy

Three-Stage Training Approach:

➤ Stage 1: Feature Extraction

The pre-trained backbone was frozen, and only the classification head was trained for 4 epochs to learn domain-specific mappings from radiographic features to class labels.

➤ Stage 2: Partial Fine-Tuning

The upper convolutional layers of the backbone were selectively unfrozen, enabling adaptation of high-level features over 8 epochs with a reduced learning rate of 1×10^{-4} .

➤ Stage 3: Full Fine-Tuning

All layers were unfrozen, and the entire model was fine-tuned for an additional 10 epochs using a learning rate of 1×10^{-5} , ensuring convergence without destabilizing previously learned representations.

8. Optimization and Loss Function

- Optimizer: Adam
- Learning rate: 1×10^{-5}
- Loss function: Categorical Cross-Entropy

Categorical Cross-Entropy is suitable for multi-class classification:

$$Loss = -\sum y_i \log(p_i)$$

9. Model Evaluation

Performance was evaluated using:

- Accuracy: The proportion of total predictions that are correct.
- Precision: The proportion of predicted positive cases that are actually positive.
- Recall: (Sensitivity): The proportion of actual positive cases correctly identified by the model
- F1-score: The harmonic mean of precision and recall, balancing both metrics.
- Confusion matrix: A table showing counts of true positives, true negatives, false positives, and false negatives.
- ROC curve: A graph plotting true positive rate against false positive rate at different classification thresholds.

These metrics provide comprehensive assessment beyond accuracy.

10. Confidence Calibration

Neural networks often produce overconfident predictions. Temperature scaling was applied to calibrate probability outputs to correct the overconfidence.

Calibrated probability:

$$p_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

where T is the temperature parameter.

Calibration improves reliability of confidence estimates. And Calibration curves are generated to assess the reliability of predicted probabilities, which is essential for clinical decision-making.

11. Explainability and Feature Analysis

❖ Occlusion Sensitivity

Occlusion sensitivity analysis is used to explain model predictions by masking regions of

the input image and observing changes in confidence scores. This method is model-agnostic and more stable for deep architectures compared to gradient-based methods. Identifies regions that significantly influence predictions by masking image patches.

❖ **t-SNE Visualization**

t-SNE is applied to high-level feature embeddings to visualize class separability in the learned feature space.

❖ **Misclassification Analysis**

Incorrect predictions are analysed to identify confusing patterns between pneumonia classes, providing insight into model limitations.

12. Confidence-Based Decision Support

The decision support system categorizes predictions based on confidence levels:

- $\geq 0.85 \rightarrow$ diagnosis accepted
- $0.60\text{--}0.85 \rightarrow$ expert review suggested
- $< 0.60 \rightarrow$ further investigation recommended

This mechanism improves clinical safety.

RESULTS AND ANALYSIS

Overview of Model Performance

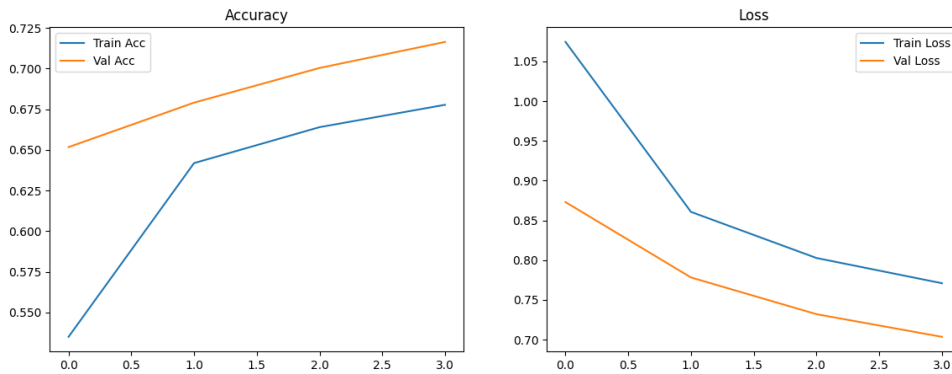
The proposed pneumonia detection framework based on EfficientNetV2-B0 was evaluated on a four-class chest X-ray dataset consisting of Normal, COVID-19, Viral Pneumonia, and Bacterial Pneumonia images. Model performance was assessed using quantitative metrics, calibration analysis, explainability techniques, and feature visualization.

The model achieved an overall classification accuracy of 82%, demonstrating effective multi-class discrimination capability.

1. Training Performance Analysis

i. Accuracy and Loss Curves

Training and validation accuracy steadily improved over epochs, while loss consistently decreased.



Observations:

- Accuracy steadily improves.
- Validation accuracy is higher than training → good generalization.
- Both training and validation loss decreased steadily.
- No significant divergence between curves.
- No overfitting observed

This indicates stable learning without severe overfitting. Training and validation accuracy increase steadily while loss decreases, indicating stable convergence. The validation accuracy slightly exceeding training accuracy suggests effective generalization and regularization through augmentation and preprocessing.

2. Classification Report Analysis

The classification performance of the proposed EfficientNetV2-B0 model was evaluated using precision, recall, and F1-score across four classes: Normal, COVID, Viral Pneumonia and Bacterial Pneumonia.

	Precision	Recall	F1-score
Normal	81%	73%	77%
Covid	81%	71%	76%
Viral	72%	86%	79%
Bacterial	94%	98%	96%

➤ **Class-wise Performance Analysis**

❖ Normal Class

- High precision indicates reliable identification.
- Lower recall suggests some normal cases predicted as pneumonia.

❖ COVID-19

- Moderate recall due to similarity with viral pneumonia.

❖ Viral Pneumonia

- High recall ensures minimal missed viral cases.
- Lower precision indicates some false positives.

❖ Bacterial Pneumonia

- Highest performance across all metrics.
- Indicates distinct radiographic features.

➤ **Overall Performance**

- Overall Accuracy: 82%
- Macro Average F1-Score: 0.82
- Weighted Average F1-Score: 0.82

This indicates balanced performance across all classes without significant bias.

➤ **Insights**

- ✓ Model performs best for bacterial pneumonia detection
- ✓ Viral and COVID cases show moderate overlap (clinically expected)
- ✓ Balanced macro and weighted averages confirm robust multi-class performance
- ✓ High recall values support clinical safety by minimizing missed pneumonia cases.

3. Confusion Matrix Analysis

The confusion matrix reveals class-wise prediction performance.

Key Observations

✓ **Bacterial Pneumonia**

- Highest correct predictions (1463)
- Very low confusion → easiest to detect

✓ **Viral Pneumonia**

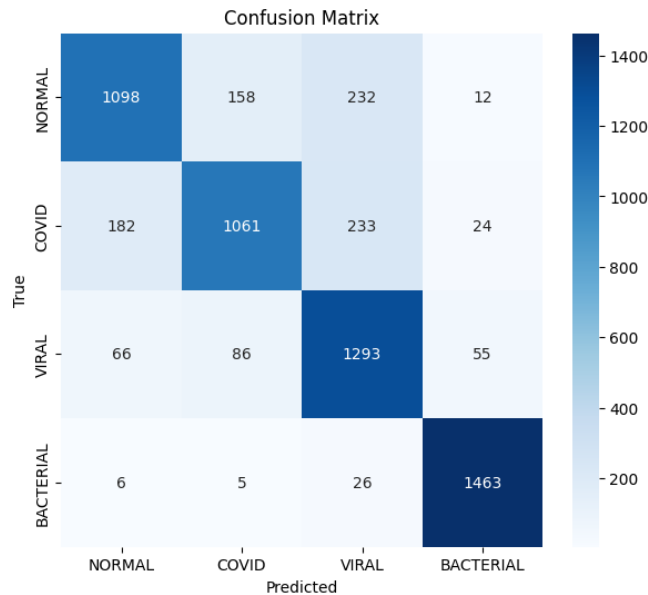
- Strong performance (1293 correct)
- Some confusion with COVID (86)

✓ COVID-19

- Confused with Viral (233) and Normal (182)
- Indicates similar radiographic patterns

✓ Normal

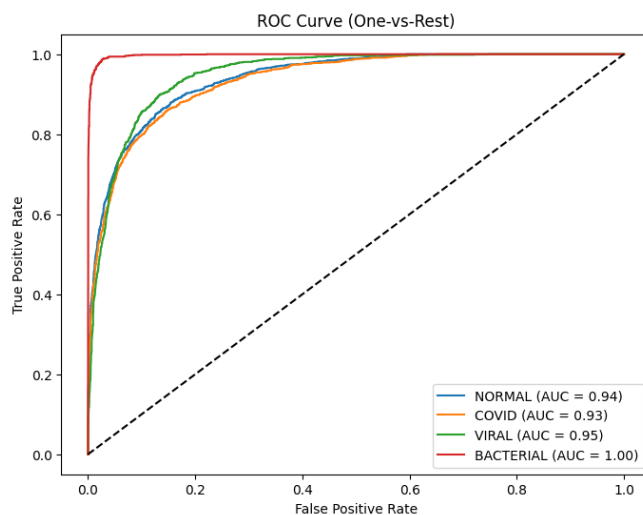
- Misclassified as Viral (232)
- Early pneumonia may resemble normal lungs



The confusion matrix demonstrates strong classification performance, particularly for bacterial pneumonia, which exhibits distinct radiographic features. Viral and COVID-19 pneumonia show moderate misclassification due to overlapping imaging characteristics. Some normal cases are incorrectly predicted as pneumonia, highlighting the challenge of distinguishing early pathological changes from healthy lung patterns.

4. ROC Curve Analysis

The One-vs-Rest ROC curves show excellent class separability.



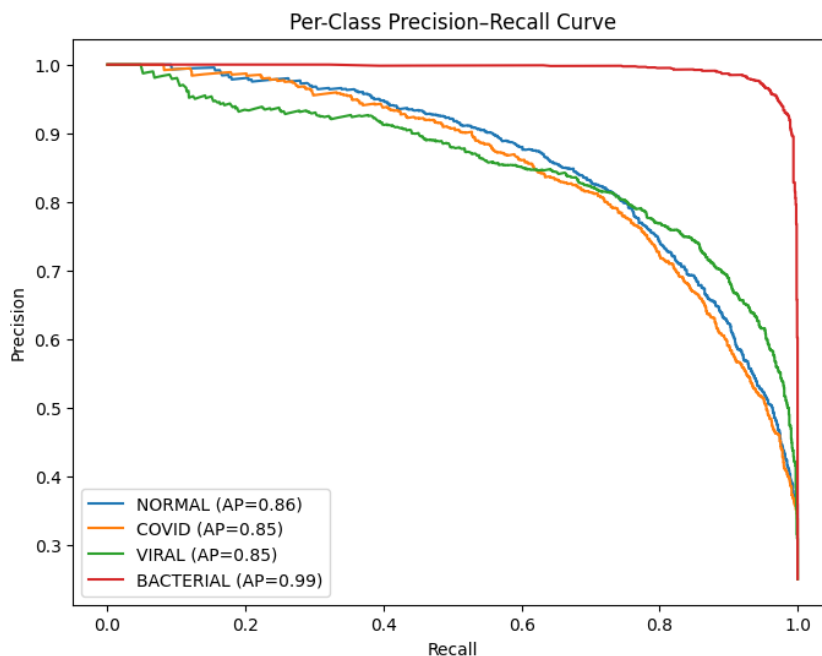
AUC Scores:

- Normal: 0.94
- COVID-19: 0.93
- Viral Pneumonia: 0.95
- Bacterial Pneumonia: 1.00

ROC analysis demonstrates excellent discriminative capability across all classes, with bacterial pneumonia achieving near-perfect separability. Slightly lower AUC values for COVID-19 reflect the similarity of viral infection patterns. High AUC values confirm robustness of the classifier.

5. Precision–Recall Curve Analysis

Precision-Recall curves provide insight into performance under class imbalance.



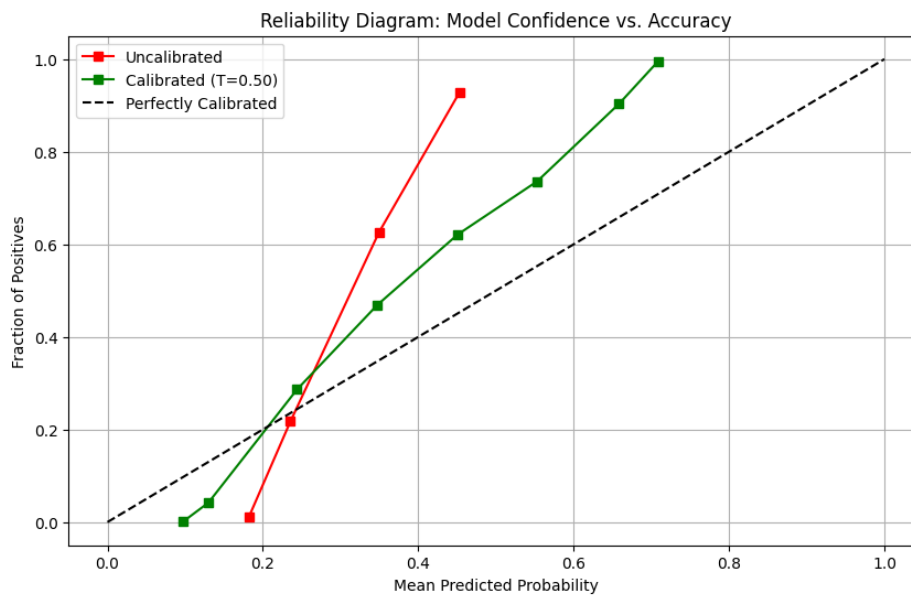
❖ **Average Precision (AP):**

- Normal: 0.86
- COVID-19: 0.85
- Viral: 0.85
- Bacterial: 0.99

The precision–recall curves indicate excellent detection performance for bacterial pneumonia, while COVID-19, viral pneumonia, and normal classes demonstrate strong but comparatively lower precision due to overlapping imaging characteristics. High AP confirms reliable detection across classes.

6. Confidence Calibration Analysis

Reliability diagrams compare predicted confidence with actual accuracy.

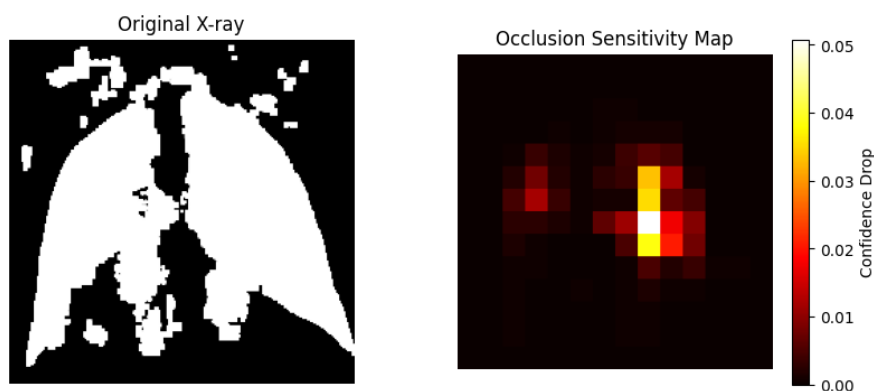


Observations:

Before calibration, predictions were overconfident. After temperature scaling, calibration curve aligns closely with the ideal diagonal. This indicates well-calibrated probability estimates, improving reliability and trustworthiness for clinical decision support.

7. Occlusion Sensitivity Analysis (Explainability)

Occlusion sensitivity maps were generated to visualize model attention.



Observations:

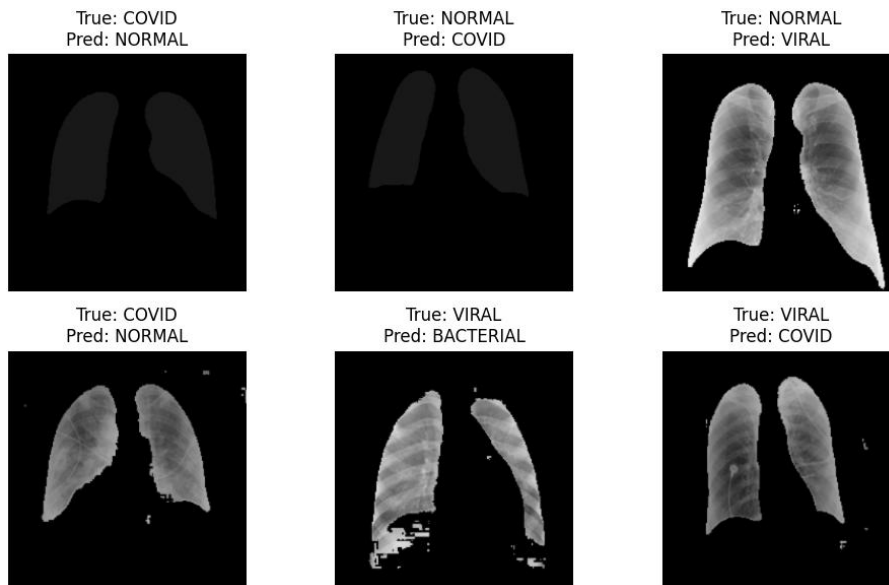
- Bright regions correspond to areas influencing predictions.
- Model focuses on lung opacities and infection-related regions.
- Non-lung regions show minimal importance.

The occlusion sensitivity map highlights critical lung regions where masking causes a significant drop in prediction confidence. This confirms the model relies on clinically relevant

regions rather than background artifacts for decision making.

8. Misclassification Analysis

Misclassified samples reveal challenging cases:

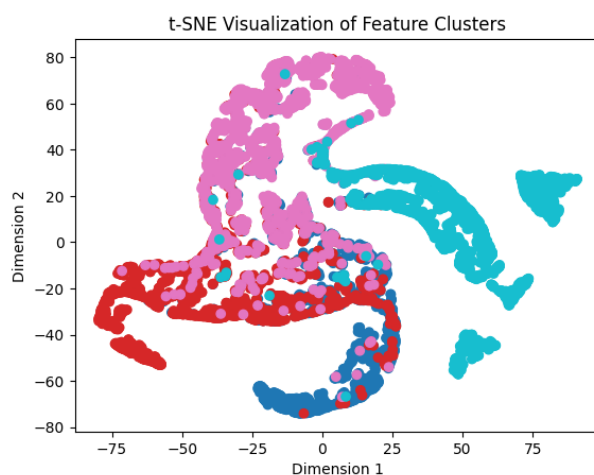


Observed Patterns:

- COVID misclassified as Normal when opacities are subtle.
- Viral pneumonia misclassified as bacterial due to overlapping features.
- Normal misclassified due to noise or imaging artifacts.

Misclassification cases highlight the inherent difficulty in distinguishing early-stage infections and overlapping pneumonia patterns. Subtle opacities and image quality variations contribute to prediction uncertainty. Errors mainly occur in borderline or low-contrast cases, highlighting challenges even for radiologists.

9. Feature Space Visualization using t-SNE



t-SNE visualization was used to examine feature embeddings. The t-SNE visualization reveals well-formed clusters, indicating effective feature learning.

Observations:

- Distinct clusters formed for each class.
- Bacterial pneumonia forms a clearly separable cluster.
- COVID and Viral clusters show partial overlap.
- Some boundary mixing indicates difficult cases.

The model learns meaningful feature representations while reflecting real clinical similarities.

COMPARISON WITH EXISTING METHODS

Deep learning–based pneumonia detection has evolved from shallow CNNs to advanced transfer learning models. Earlier works such as CheXNet (Rajpurkar et al., 2017) demonstrated radiologist-level pneumonia detection using DenseNet121 but focused primarily on binary classification. The base paper (Ibrahim et al., 2021) employed AlexNet for multi-class pneumonia detection; however, its shallow architecture limited feature extraction capability.

Recent studies using EfficientNet architectures have shown improved performance due to compound scaling and better feature reuse. For example, EfficientNet-based models in medical imaging achieved higher accuracy and improved convergence stability. CLAHE-based enhancement methods (Altan & Narlı, 2022) further demonstrated that preprocessing improves diagnostic feature visibility.

Compared to these approaches, the proposed system integrates:

- EfficientNetV2-B0, a modern, efficient architecture
- Image enhancement to improve visibility of infection patterns
- U-Net lung segmentation to focus analysis on lung regions
- Confidence calibration for reliable probability estimates
- Explainability techniques (occlusion sensitivity)
- Feature visualization for model interpretability

While traditional methods emphasize classification accuracy, the proposed framework prioritizes clinical reliability, interpretability, and robustness, making it more suitable for real-world deployment.

ADVANTAGES OF THE PROPOSED WORK

1. **Improved Feature Learning**
EfficientNetV2-B0 extracts richer and more discriminative features compared to older CNN architectures.
2. **Enhanced Image Quality**
Contrast and brightness enhancement improves visibility of subtle pneumonia patterns such as opacities and infiltrates.
3. **Region-Focused Analysis**
U-Net lung segmentation removes background noise and ensures the model focuses on clinically relevant lung regions.
4. **Reliable Predictions**
Confidence calibration produces well-calibrated probability scores, improving trust in model predictions.
5. **Explainability & Transparency**
Occlusion sensitivity maps provide visual evidence of decision regions, supporting clinical interpretability.
6. **Robust Multi-Class Detection**
The model successfully distinguishes between Normal, COVID-19, Viral, and Bacterial pneumonia.
7. **Strong Class Separability**
ROC and Precision-Recall analyses confirm high discrimination ability.
8. **Generalization Capability**
Balanced performance across classes demonstrates model robustness.
9. **Clinical Decision Support Readiness**
Confidence-based decision logic supports risk-aware diagnosis.

LIMITATIONS

Despite promising performance, several limitations exist:

1. **Moderate Confusion Between COVID and Viral Pneumonia**
Radiographic similarities make differentiation challenging.
2. **Sensitivity to Image Quality**
Low-contrast or noisy X-rays can reduce detection accuracy.
3. **Segmentation Dependency**
Errors in lung segmentation may affect classification performance.
4. **Limited Dataset Diversity**
Performance may vary across hospitals or imaging devices.
5. **Computational Cost**
Deep learning models require GPU resources for efficient training.
6. **Lack of Clinical Metadata Integration**
Patient history and symptoms were not incorporated.
7. **Explainability Still Approximate**
Occlusion maps highlight important regions but do not provide causal explanations.

FUTURE ENHANCEMENTS

Future improvements can further enhance the system:

1. **Lesion Localization Integration**
Incorporating lesion segmentation can improve disease-specific focus.
2. **Attention Mechanisms**
Attention modules can help the model focus on infection patterns.
3. **Multi-Modal Data Integration**
Combining clinical reports, lab results, and symptoms may improve diagnosis accuracy.
4. **Advanced Explainability Methods**
Grad-CAM++, SHAP, or saliency maps can provide deeper interpretability.
5. **Deployment Optimization**
Model compression and quantization can enable real-time deployment.
6. **Larger Multi-Center Dataset Training**
Training on diverse datasets improves generalization.
7. **Self-Supervised Pretraining**
Leveraging unlabeled medical images may enhance feature learning.
8. **Edge Deployment for Rural Healthcare**
Lightweight versions can support portable diagnostic tools.

CONCLUSION

This work presents an enhanced deep learning framework for automated pneumonia detection using chest X-ray images. By integrating EfficientNetV2-B0, image enhancement techniques, lung segmentation, confidence calibration, and explainability methods, the proposed system improves diagnostic reliability and clinical interpretability.

Experimental results demonstrate strong classification performance, robust feature learning, and reliable probability estimation. The integration of segmentation and explainability enhances trust and transparency, making the system suitable for clinical decision support.

Although challenges remain in differentiating visually similar pneumonia types and handling low-quality images, the proposed framework establishes a strong foundation for future AI-assisted respiratory disease diagnosis systems.

REFERENCES

1. A. U. Ibrahim et al., “Pneumonia classification using deep learning from chest X-ray images during COVID-19,” *Cognitive Computation*, Springer, 2021.<https://link.springer.com/article/10.1007/s12559-020-09787-5>
2. M. Tan and Q. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” *ICML*, 2019. <https://arxiv.org/abs/1905.11946>
3. P. Rajpurkar et al., “CheXNet: Radiologist-level pneumonia detection on chest X-rays,” <https://arxiv.org/abs/1711.05225>
4. S. Banerjee and M. Monir, “EfficientNet V1–V2 for medical image classification,” *Springer*, 2024. https://doi.org/10.1007/978-3-031-53717-2_30
5. S. HaghaniFar et al., “COVID-CXNet: Detecting COVID-19 in chest X-ray images,” *Multimedia Tools and Applications*, 2022. <https://doi.org/10.1007/s11042-022-12156-z>
6. M. E. Chowdhury et al., “Can AI help in screening viral and COVID-19 pneumonia?” <https://arxiv.org/abs/2003.13145>
7. G. Altan and S. S. Narlı, “CLAHE based enhancement to transfer learning in COVID-19 detection,” <https://dergipark.org.tr/en/pub/gmbd/article/1031619>
8. D. A. Moses, “Deep learning applied to automatic disease detection using chest X-rays,” *J. Med. Imaging Radiat. Oncol.*, vol. 65, no. 5, pp. 498–517, Aug. 2021 <https://onlinelibrary.wiley.com/doi/10.1111/1754-9485.13273>
9. Guo. et. al. (2017), “On Calibration of Modern Neural Networks”
10. . Karen Simonyan, Andrea Vedaldi & Andrew Zisserman (2014),” Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps” <https://arxiv.org/abs/1312.6034>
11. Naqvi. et. al . (2022), “Improved lung segmentation based on U-Net architecture and morphological operations”, <https://arxiv.org/abs/2210.10545>
12. Ronneberger et al., (2015), “U-Net: Convolutional Networks for Biomedical Image Segmentation”, <https://arxiv.org/abs/1505.04597>
13. <https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database>
14. <https://www.geeksforgeeks.org/computer-vision/efficientnet-architecture/>